

Research focuses on developing practical experimental pipelines and predictive systems for scientific and healthcare applications, integrating data-efficient machine learning, statistical inference, experimental design, and adaptive learning. Contributions include noise-aware learning systems, adaptive data acquisition, and label-efficient hypothesis testing.

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Memphis, TN 38163 https://wayne0908.github.io

RESEARCH INTERESTS Data-Efficient Machine Learning; Experimental Design; Hypothesis Testing;
AI Reliability; ML/AI for Science

PROFESSIONAL EXPERIENCE **Postdoctoral Researcher** Jul '25 – Now
Department of Genetics, Genomics and Informatics,
The University of Tennessee Health Science Center

Postdoctoral Researcher Mar '24 – Apr '25
Statistical Sciences Group, Los Alamos National Laboratory

Postdoctoral Researcher Jan '23 – Feb '24
College of Health Solutions, Arizona State University

Machine Learning Engineer Intern & Research Scientist May – Aug '21
Sep '22 – Jan '23
Meta

EDUCATION **Ph.D., Computer Engineering** Aug '18 – Aug '22
Arizona State University, Tempe, USA
Dissertation: *"Modeling and Exploiting the Structure of Data via Meta-Features for Robust and Efficient Machine Learning"*
Advisors: Dr. Visar Berisha and Dr. Gautam Dasarathy

M.S., Electrical Engineering Aug '15 – Dec '17
Texas A&M University, College Station, USA
Thesis: *"Noise-Tolerant Deep Learning for Histological Image Segmentation"*
Advisors: Dr. Jim Ji and Dr. Xiaoning Qian

B.S., Electronic Information Science and Technology Sep '11 – Jun '15
Shandong University, China
Outstanding Undergraduate Thesis: *"Influences of Two Image Dehazing Filters on Image Compression Algorithms"*

CONFERENCE PAPERS

1. **Li, Weizhi**, Xiaoning Qian, and Jim Ji. "Noise-tolerant deep learning for histopathological image segmentation." In 2017 IEEE International Conference on Image Processing (ICIP), pp. 3075-3079. IEEE, 2017.
2. **Li, Weizhi**, Gautam Dasarathy, and Visar Berisha. "Regularization via structural label smoothing." In International Conference on Artificial Intelligence and Statistics (AISTATS), pp. 1453-1463. PMLR, 2020.
3. **Li, Weizhi**, Gautam Dasarathy, Karthikeyan Natesan Ramamurthy, and Visar Berisha. "Finding the homology of decision boundaries with active learning." Advances in Neural Information Processing Systems 33 (2020): 8355-8365.
4. **Li, Weizhi**, Gautam Dasarathy, Karthikeyan Natesan Ramamurthy, and Visar Berisha. "A label efficient two-sample test." In Uncertainty in Artificial Intelligence, pp. 1168-1177. PMLR, 2022.
5. **Li, Weizhi**, Visar Berisha, and Gautam Dasarathy. "Advanced Tutorial: Label-Efficient Two-Sample Tests." In 2024 Winter Simulation Conference (WSC), pp. 131-145. IEEE, 2024.

6. Zahin, Abrar, **Weizhi Li**, and Gautam Dasarathy. "Rapid Change Localization in Dynamic Graphical Models." In ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 7770-7774. IEEE, 2024.

7. Dolin, Pavel, **Weizhi Li**, Gautam Dasarathy, and Visar Berisha. "Statistically Valid Post-Deployment Monitoring Should Be Standard for AI-Based Digital Health." In The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

JOURNAL PAPERS

1. Wang, Liping, Xiao Zhou, Chengyou Wang, and **Weizhi Li**. "The Effects of Image Dehazing Methods Using Dehazing Contrast-Enhancement Filters on Image Compression." KSII Transactions on Internet and Information Systems (TIIS) 10, no. 7 (2016): 3245-3271.

2. Tsai, Chung-Chi, **Weizhi Li**, Kuang-Jui Hsu, Xiaoning Qian, and Yen-Yu Lin. "Image co-saliency detection and co-segmentation via progressive joint optimization." IEEE Transactions on Image Processing 28, no. 1 (2018): 56-71.

3. **Li, Weizhi**, Prad Kadambi, Pouria Saidi, Karthikeyan Natesan Ramamurthy, Gautam Dasarathy, and Visar Berisha. "Active Sequential Two-Sample Testing." Transactions on Machine Learning Research.

4. **Li, Weizhi**, Klein, N., Gifford, B., Sklute, E., Legett, C., & Clegg, S. M. (2026). Regularization via f-divergence: An application to multi-oxide spectroscopic analysis. Journal of Geophysical Research: Machine Learning and Computation, 3, e2025JH000990. <https://doi.org/10.1029/2025JH000990>

ABSTRACTS

1. **Li, Weizhi**, Natalie Klein, and Samuel M. Clegg. "Enhancing Neural Network Generalizability with f-Divergence Regularization for Multi-Oxide Spectroscopic Analysis." In AGU Fall Meeting Abstracts, vol. 2024, pp. V13F-04. 2024.

2. **Li, Weizhi**, D. Das, E. Sklute, C. Legett, N. Klein, and S. Clegg. "Multi-Oxide Analysis Through the Fusion of LIBS and VISIR Spectroscopic Data." LPI Contributions 3090 (2025): 1349.

3. Das, D., R. Martinez, C. Legett, E. Sklute, S. Clegg, A. Ziemann, U. Wolf, and **W. Li**. "Building and Characterizing a VISIR Reflectance Library and Spectral Database at Los Alamos National Laboratory." LPI Contributions 3090 (2025): 2095.

4. M. Hinds, **W. Li**, et al., *Explainable Deep Learning for Laser Induced Breakdown Spectroscopy*. **Honorable Mention**. Conference on Data Analysis 2025

WORKING PAPERS

1. Hinds, Mark, Michael Geyer, Weizhi Li, Elizabeth Sklute, Debarati Das, Ronald K. Martinez, Allison Zastrow et al. "A Comparative Study of Multimodal Data Fusion Strategies for Planetary Spectroscopy." Available at SSRN 6439537.

2. **Li, Weizhi**, Gautam Dasarathy, and Visar Berisha. "Matched-Pair Experimental Design with Active Learning." arXiv preprint arXiv:2509.10742 (2025).

SOFTWARE

1. Klein, Natalie, and **Weizhi Li**. "NNFDivergence." US Department of Energy (DOE) Software (2025): 149.

GRANTS

1. NIH R01 (NLM), Submitted, 2026. In-Context Learning for Robust Omics-Based Disease Prediction Across the Genetic Ancestry Continuum. Status: Under review.

TALKS & PRESENTATIONS

Invited Talks

Invited Talk: *Statistics Seminar*, Department of Mathematics & Statistics, University of New Mexico

Topic: *Label-Efficient Two-Sample Tests*

Mar '25

Tutorial talk: *Winter Simulation Conference*, Orlando, Florida

Topic: *Label-Efficient Two-Sample Tests*

Dec '24

Invited talk: *Applied Machine Learning Summer School*, Los Alamos National

	Laboratory	
	Topic: <i>AI-Enhanced Exploration for Planetary Science</i>	July '24
	Invited talk: <i>AI4OPT and AI4Science Reading Group</i> , Los Alamos National Laboratory	
	Lead a Discussion on <i>Physics-Based Bayesian Optimization</i>	Apr '24
	Invited talk: <i>LIONS Seminar Series</i> , Arizona State University	
	Topic: <i>Designing Two-Sample Tests with Local Information</i>	Sep '23
	Conference Presentations	
	Oral Presentation: <i>American Geophysical Union (AGU)</i> , Washington, D.C.	Dec '24
	Poster presentation: <i>Asilomar Conference on Signals, Systems, and Computers</i> , Pacific Grove, CA	Oct '22
	Invited talk: <i>Journal Club on Responsible Use of Machine Learning Research</i> , ECEE Department at ASU, Tempe, AZ	Sep '22
	Poster presentation: <i>Conference on Uncertainty in Artificial Intelligence</i> , Virtual	Aug '22
	Poster presentation: <i>Conference on Neural Information Processing Systems (NeurIPS) Systems</i> , Virtual	Dec '20
	Poster presentation: <i>International Conference on Artificial Intelligence and Statistics</i> , Virtual	Aug '20
	Poster presentation: <i>International Conference on Image Processing</i> , Beijing, China	Sep '17
SUPERVISED STUDENTS	Robert Lattus (Student at ASU, now pursuing Ph.D. at University of Florida) Mark Hinds (Student at LANL)	
AWARDS	NeurIPS Top Reviewer Award	Oct '25
	UAI Student scholarship	Jun '22
	Graduate Travel Award from Arizona State University (Remote)	Aug '20
	Engineering Graduate Fellowship, Arizona State University,	2018–2019
	Poster Award in SWE Region C Conference Poster Competition	Mar '17
	Graduate Merit Scholarship from Texas A&M University	Aug '16
	Shandong University Departmental Scholarship	Oct '14
SERVICES & ACTIVITIES	Journal Reviewer for Transactions on Information Theory, Journal of Combinatorial Optimization	
	Conference Reviewer for NeurIPS, ICML, ICLR, UAI, IJCAI, AAAI	
	Mentor for DataFest hackathon hosted in Arizona State University	Mar '23
	Student panel at <i>Ph.D. open house</i> for the School of Computing and Augmented Intelligence at ASU	Feb '22
	Graduate Fulton Ambassador in Arizona State University	May – Nov '20
	Medical Imaging Summer School: Medical Imaging Meets Machine Learning [Activity Link]	Aug '16